

Understanding the factors behind customer dissatisfaction and strategies to mitigate risks in the fashion e-commerce sector.

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Delivery failures are predictable—and put ~R\$600K in customer value at risk

Customer dissatisfaction due to delivery failures is not random; it is concentrated in specific scenarios that can be anticipated and mitigated. This signifies a substantial risk to revenue and brand integrity, highlighting the urgent need for enhanced operational insights.





Reactive Feedback Mechanisms

This approach can lead to a delayed response to issues that could have been mitigated earlier in the process.



Bias in Customer Feedback

Feedback often skews towards extremes, leading to a biased view of customer sentiment and limiting understanding of broader patterns.



Need for Proactive Insights

By focusing on predictive insights, companies can address issues in real-time, reducing dissatisfaction and enhancing the overall customer journey.



Customer dissatisfaction is typically addressed too late

Fast, reliable delivery is a baseline expectation

- In today's retail environment, consumers demand not only quality products but also outstanding order fulfillment and personalization.



Delivery timing directly affects product relevance

- In the fashion industry, the relevance of a product can diminish rapidly due to trends and seasonal changes.
- Timely delivery ensures that customers receive their desired items when they are most likely to be worn and appreciated.



High return rates are linked to post-purchase failure

- **Returns** often arise from unmet expectations in the post-purchase experience – which usually **begins with a poor delivery scenario**.
- In an industry already under pressure to reduce waste, avoidable returns and re-shipments add another layer of inefficiency.

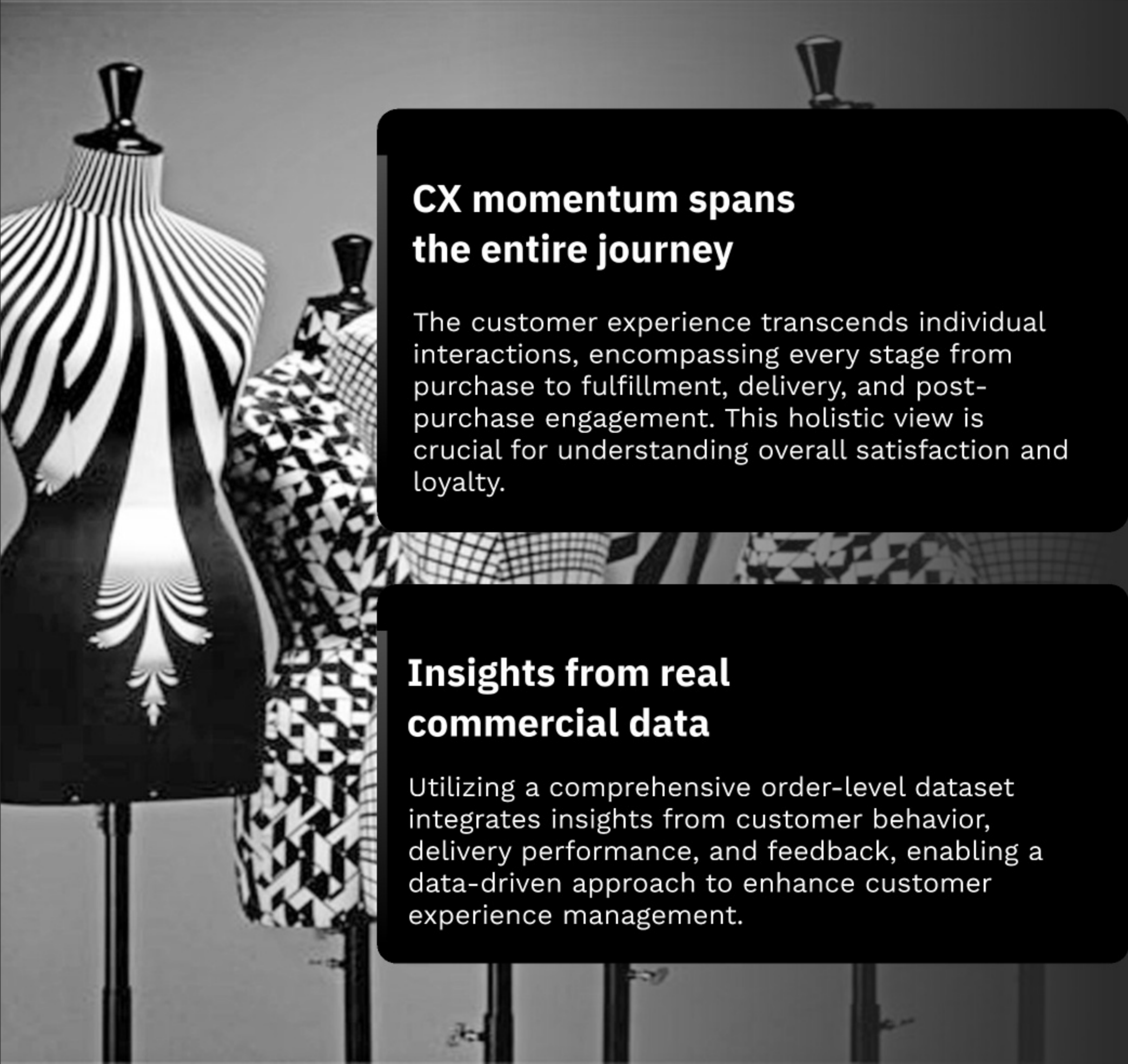


**Customer
experience breaks
when execution
fails the promise**

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CX momentum spans the entire journey

The customer experience transcends individual interactions, encompassing every stage from purchase to fulfillment, delivery, and post-purchase engagement. This holistic view is crucial for understanding overall satisfaction and loyalty.

Insights from real commercial data

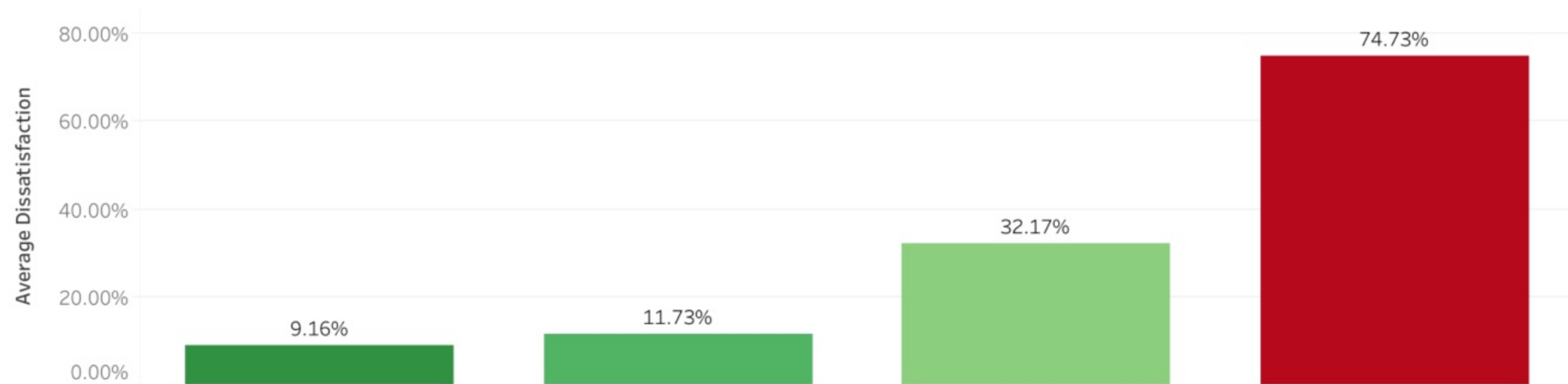
Utilizing a comprehensive order-level dataset integrates insights from customer behavior, delivery performance, and feedback, enabling a data-driven approach to enhance customer experience management.

Customer experience is a journey—not a single event

Delivery Performance Drives Dissatisfaction

Delivery performance is the clearest operational driver of dissatisfaction. While most orders are delivered on time or early, dissatisfaction rises sharply once expectations are missed.

Dissatisfaction spikes when deliveries are late



Most orders arrive on or before the estimated delivery date



Observed dissatisfaction varies across months



Business Exposure

75%

is how high
dissatisfaction
rates can get when
orders are late.

R\$ 763K

worth of late
orders

R\$590K

associated with
dissatisfied
customers

Actionable Scenario

Reducing late deliveries by 20% would:

~845

late orders
prevented

~636

negative
experiences avoided

~R\$118K

Safeguarded in
reduced exposure

From diagnosing failures → enabling operational decisions

Leveraging predictive insights to enhance operational effectiveness and customer satisfaction.



Model Development

Implemented a supervised learning model integrating delivery, order, and customer features.



Performance Optimization

Gradient boosting demonstrated superior accuracy in predicting customer dissatisfaction.



Evaluation Metrics

Utilized ROC-AUC, Recall, and F1-score to validate model effectiveness.



Focus on Recall

Prioritized recall to ensure high-risk orders are captured and addressed.



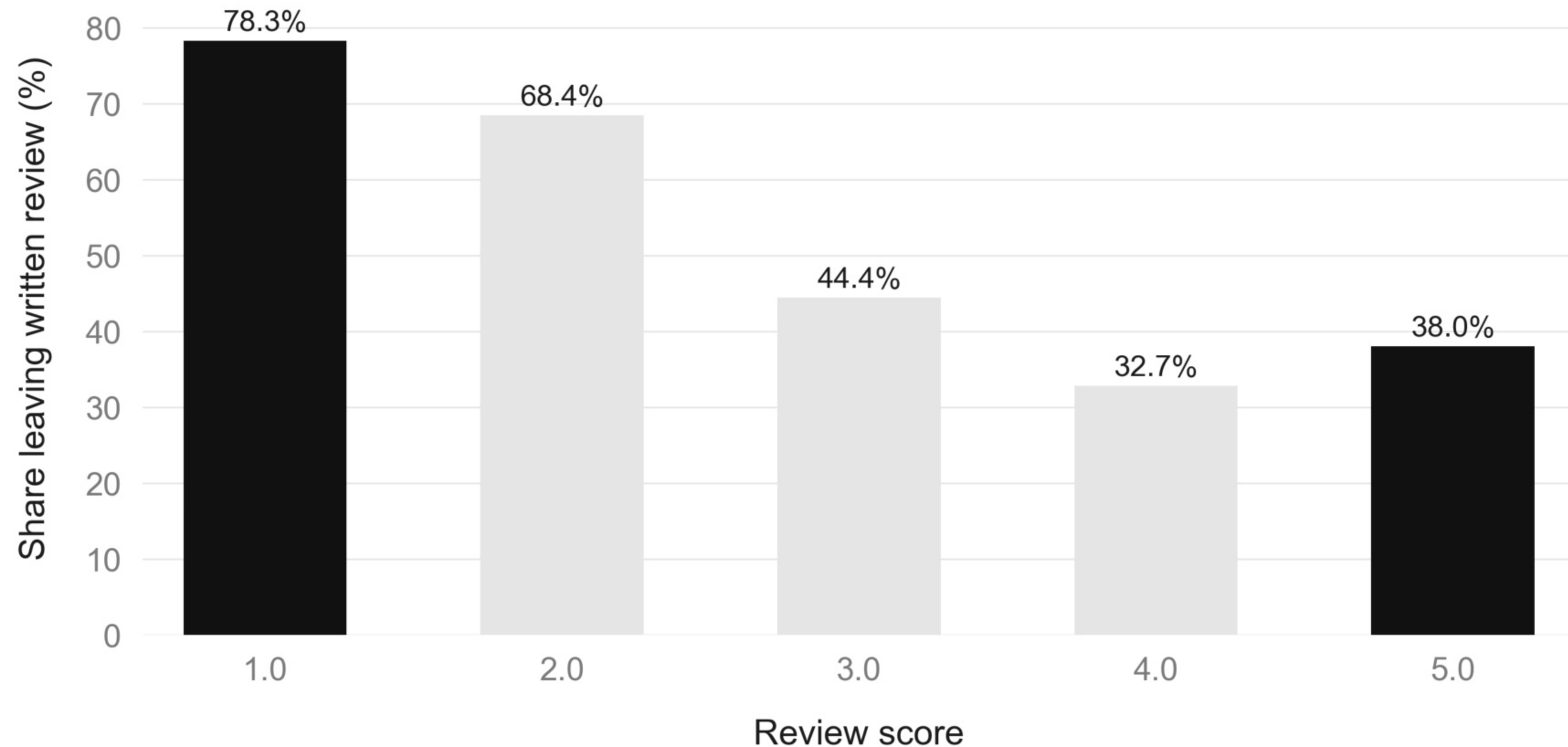
Key Predictors

Identified delivery-related variables as the strongest indicators of potential failure.



Customers are more likely to leave written reviews at the extremes

Both very negative and very positive experiences trigger written feedback

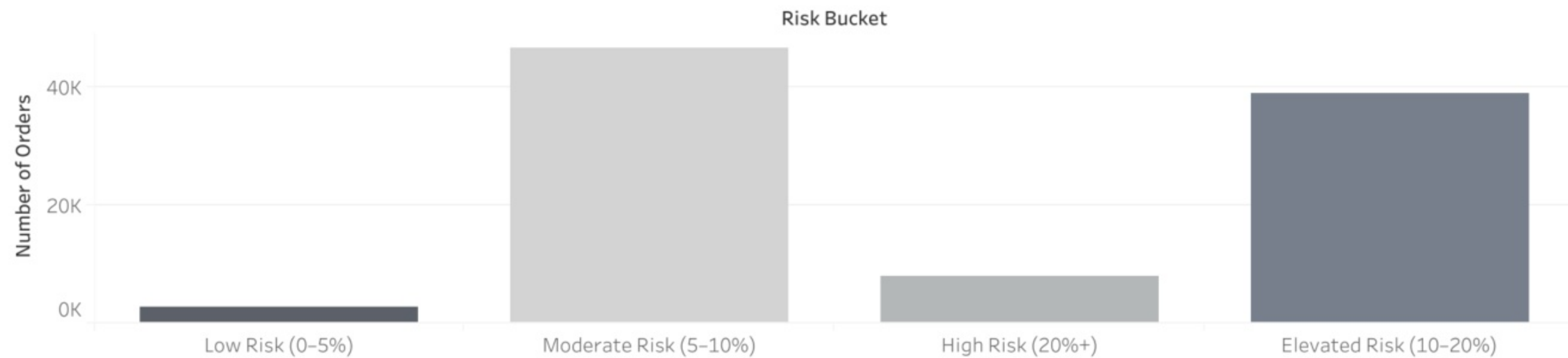


	Model	ROC-AUC	Recall (Dissatisfied)	F1 (Dissatisfied)
2	Gradient Boosting	0.735237	0.276923	0.402946
1	Random Forest	0.734790	0.277328	0.404011
0	Logistic Regression	0.729352	0.495142	0.448643

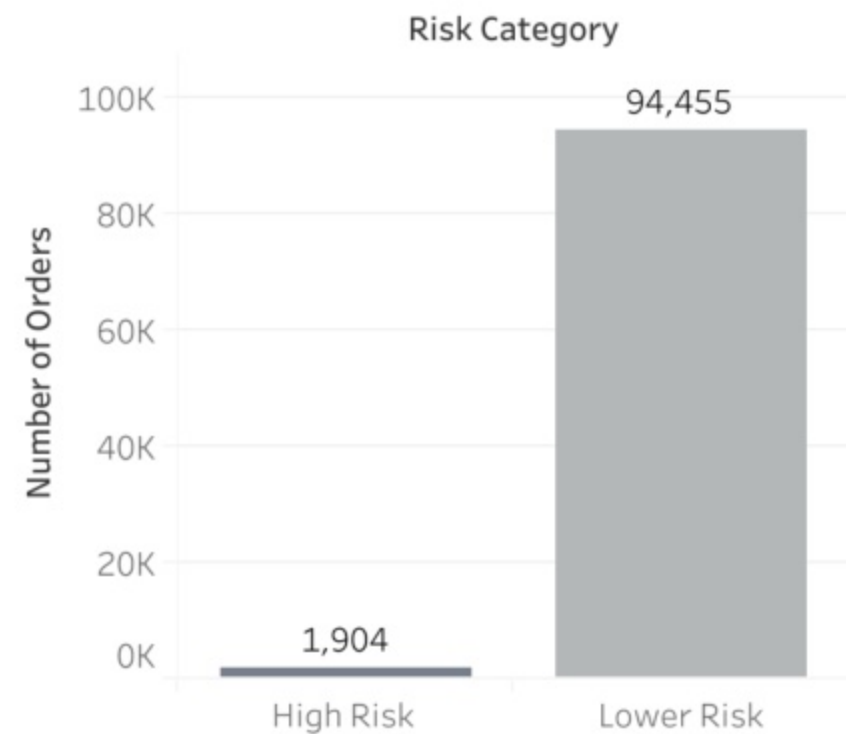
Identifying At-Risk Orders Before Complaints Surface

The model extends the analysis from observed dissatisfaction to predicted dissatisfaction risk, allowing at-risk orders to be identified before negative feedback is expressed.

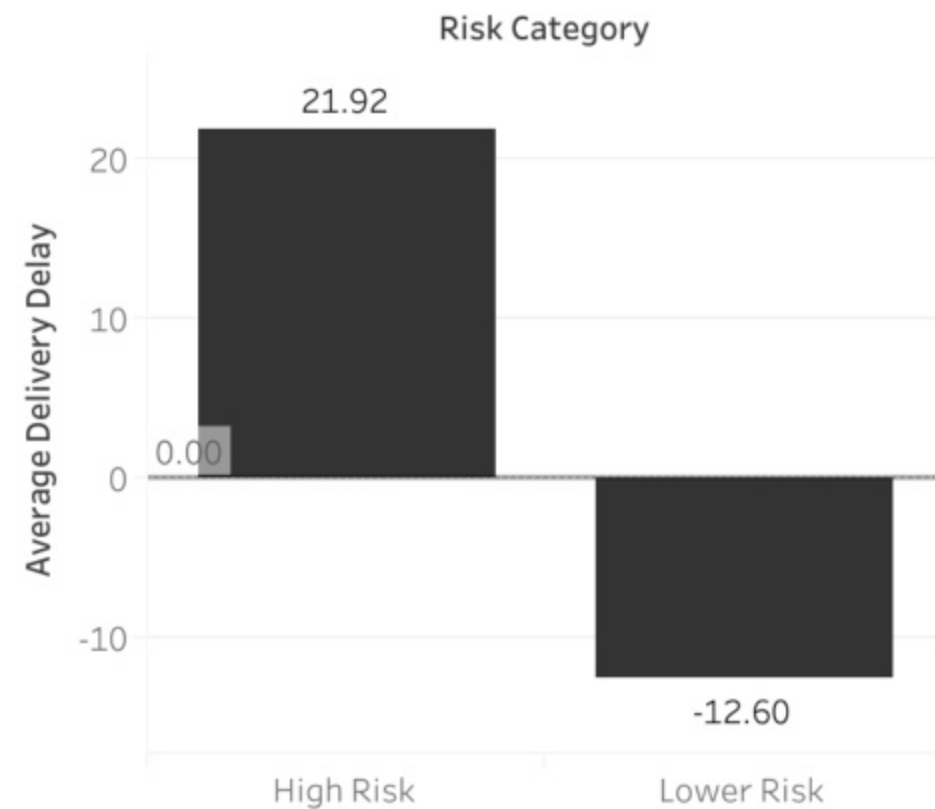
Most orders are low-risk, but a meaningful segment carries elevated dissatisfaction risk



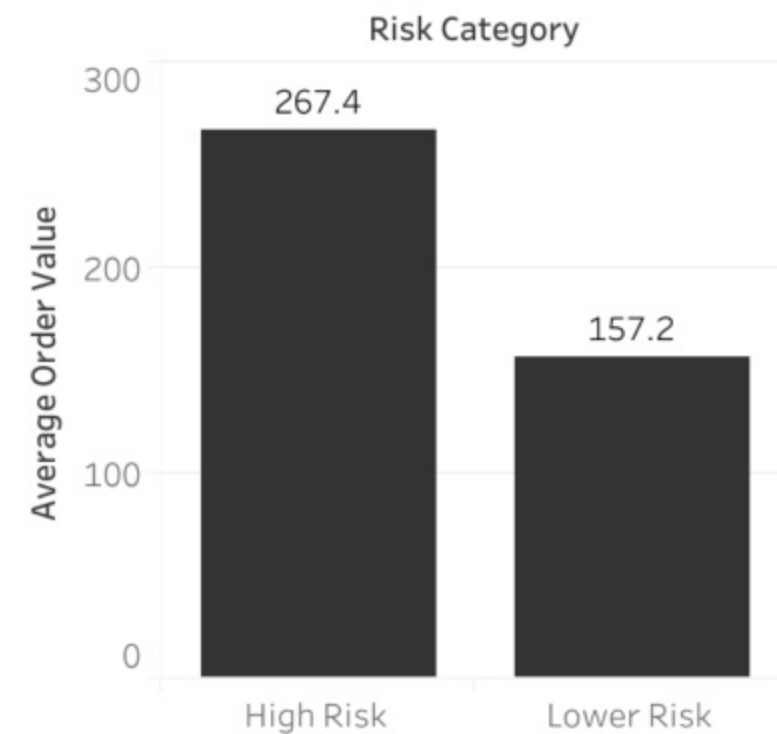
A subset of orders can be flagged for proactive intervention



High-risk orders show weaker delivery performance



High-risk orders can also carry meaningful commercial value



**Preventing experience
failure requires operating
model alignment**



Proactively identify high-risk orders



Optimize fulfillment for at-risk shipments



Enhance proactive customer communication



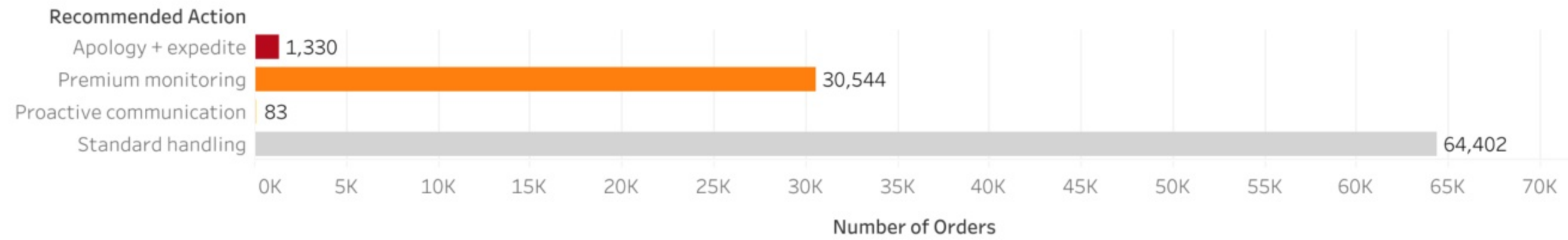
Integrate logistics and customer experience



From Prediction to Action

This transforms this project from descriptive analysis into a decision-support tool. Instead of reacting only after dissatisfaction..

How the system would prioritize interventions



Which high-risk orders matter most commercially



In fashion, delivery is not logistics—it is part of the product

Ship-from-Store Decision System:

1. Real-time Data:

- Store inventory
- Warehouse stock
- Customer orders
- Store capacity
- Carrier performance
- Shipping costs

2. System Setup:

- Order Management System
- Inventory Management System
- Fulfillment engine
- APIs

3. Optimization Goals:

- Low costs
- Fast delivery
- Minimize CX risk



Instead of treating all orders equally, companies can prioritize:

- high-risk orders
- high-value orders
- and time-sensitive deliveries

Key Insights

- E-commerce dissatisfaction isn't random.
- It stems from delivery performance, which can be tracked and predicted.
- We can turn these insights into actions for better order fulfillment.
- Our main aim is to not just understand customer experiences but to create systems that safeguard them.

**From reactive
service to
proactive
experience
management**

Data Preparation

Built a comprehensive order-level dataset integrating orders, customer profiles, reviews, and delivery metrics to capture the full customer journey.

Exploratory Analysis

Analyzed reviewed orders to pinpoint critical factors impacting customer satisfaction, focusing particularly on delivery delays and expectation management.

Modeling

Employed advanced modeling techniques including Logistic Regression, Random Forest, and Gradient Boosting, optimizing performance through hyperparameter tuning.

Evaluation

Evaluated model performance utilizing metrics such as ROC-AUC, Recall, and F1-score, prioritizing Recall to effectively capture high-risk scenarios.

Key Insight

Key findings revealed that delivery-related features are the dominant predictors of customer dissatisfaction, affirming the operational nature of the problem.

Appendix: Technical Approach & Model Overview

Take this with you. Revisit anytime.

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